Burned Area Intensity Mapping using Prithvi

PRYANJALI GOEL and MARIA KAPUSHEVA, Leiden University, The Netherlands

CCS Concepts: • **Do Not Use This Code → Generate the Correct Terms for Your Paper**; *Generate the Correct Terms for Your Paper*; Generate the Correct Terms for Your Paper; Generate the Correct Terms for Your Paper.

Additional Key Words and Phrases: Do, Not, Us, This, Code, Put, the, Correct, Terms, for, Your, Paper

ACM Reference Format:

Pryanjali Goel and Maria Kapusheva. 2018. Burned Area Intensity Mapping using Prithvi. 1, 1 (June 2018), 5 pages. https://doi.org/XXXXXXX.XXXXXXX

1 Introduction

The increasing frequency and severity of wildfires, driven by the ongoing climate crisis, presents a critical challenge for environmental monitoring and disaster management. These intensifying fire events necessitate advanced methods for accurate assessment of burned areas and their severity. While the mapping of burned areas serves essential functions in forestry management, damage assessment, and fire prediction modeling, current methods often lack the capability to quantify burn intensity—a crucial parameter for understanding fire impacts and planning recovery efforts. Recent advances in deep learning approaches have shown promising results in burned area mapping. Notably, in the work of pinto2021practical demonstrated the effectiveness of a multi-step Convolutional Neural Network (CNN) architecture in accurately and efficiently identifying burned regions from satellite imagery. However, there remains significant potential to extend this work by incorporating burn intensity assessment capabilities, which would provide more comprehensive information for both immediate response and long-term forest management strategies. The emergence of large-scale transformer-based models specifically designed for satellite imagery analysis, such as Prithvi Prithvi-100M, opens new possibilities in this domain. These foundation models have demonstrated remarkable capabilities in understanding complex geospatial patterns and features across various remote sensing applications. This research explores whether Prithvi's sophisticated architecture can be effectively fine-tuned to not only detect burned areas but also accurately classify their burn intensity levels. This study aims to address the following research question: Can the geospatial foundation model Prithvi be successfully fine-tuned to achieve high accuracy in mapping burn intensity from satellite imagery? By investigating this question, we seek to advance the field of wildfire damage assessment and contribute to more effective forest management and fire response strategies.

This research addresses the challenge of quantifying burn intensity in wildfire-affected areas through a comparative analysis framework. While the existing Prithvi model (available on Hugging Face) demonstrates capabilities in burn

Authors' Contact Information: Pryanjali Goel; Maria Kapusheva, Leiden University, Leiden, The Netherlands.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

© 2018 Copyright held by the owner/author(s). Publication rights licensed to ACM.

Manuscript submitted to ACM

scar segmentation, as shown in the baseline results, our study aims to extend this functionality to include burn intensity classification. The performance differential between these approaches can be quantified as:

$$\Delta Accuracy = Accuracy_{BurnIntensity} - Accuracy_{BurnScar} \tag{1}$$

where $Accuracy_{BurnIntensity}$ represents our fine-tuned model's performance in classifying burn intensity levels, and $Accuracy_{BurnScar}$ denotes the baseline Prithvi model's performance in binary burn scar segmentation. This comparison framework allows us to:

- Establish a clear benchmark using the existing Prithvi burn scar segmentation model
- Evaluate the effectiveness of our fine-tuning approach in capturing the additional complexity of burn intensity classification
- Quantify the potential trade-offs between binary segmentation and multi-class intensity classification

The core challenge lies in maintaining or improving upon the baseline segmentation accuracy while introducing the additional capability of intensity classification, which requires more nuanced feature recognition and classification boundaries.

2 Related work

Find and add related work for: Semantic segmentation Burn mapping Prithvi's burn scar Burn intensity

When approaching the task fo burn intensity mapping, it is crucial to understand what a semantic segmentation task is. Semantic segmentation tasks in satellite imagery involve the pixel-wise classification of geospatial features, where each pixel in the satellite image is assigned to a specific land cover or environmental condition class singh2023semantic. In the context of Earth observation, these tasks are particularly challenging due to the high spectral dimensionality of satellite data (multiple spectral bands), varying spatial resolutions, and complex temporal dynamics. Common applications include land use classification, urban area mapping, and natural disaster impact assessment, where models must learn to distinguish between different surface characteristics based on their spectral signatures across multiple wavelengths. In wildfire analysis specifically, semantic segmentation enables the detailed mapping of burned areas by processing multiple spectral bands simultaneously to identify subtle variations in surface reflectance that indicate fire damage.

Most of those applications involve burn area mapping. This is an example of a binary semantic segmentation task, where each pixel is labeled as burned or not burned. This is also discussed in the work of pinto2021practical, which is the original paper that inspired our project. There they focus on providing a fast method to map burned areas in high-resolution without sacrificing efficiency.

On the other hand, one of the most notable applications of the transformer-based model for satellite imagery Prithvi Prithvi-100M-preprint, is burn scar mapping, which is essentially mapping out the areas that have been damaged the most by a wildfire. WHat is important to know about this model however is that it has been trained on Harmonized LandSat(HLS) satellite images. HLS images are standardized satellite data products that combine and align observations from both Landsat and Sentinel-2 satellites into a single, consistent format with matched spatial resolution, coordinate system, and spectral bandwidths to enable seamless analysis across both data sources. In addition to that during the training process, Prithvi is levareging both the geospatial as well as the temporal information of the data. This is important, since the pre-trained model that we will use in this project can efficiently use that temporal knowledge to help with noise reduction and faster convergence.

Regarding research that focuses on burned intensity mapping, we found few works that delved into our specific scenario. Most notably lee2023evaluating investigate the applicability of different machine learning methods to the problem, however they mostly focus on the comparison with the damage scale provided by Copernicus Emergency Management system which is very limited.

3 Methods

Our approach leverages the pre-trained Prithvi model, which we fine-tuned specifically for burn intensity mapping. The model processes HLS imagery containing six spectral bands: Red, Green, Blue, Narrow NIR, SWIR1, and SWIR2. For each input image, we provide a corresponding single-band grading mask containing burn intensity classifications ranging from 0 to 4, where 0 represents unburned areas and 4 indicates severely burned regions. The dataset comprises 296 image-mask pairs, which we partitioned using an 80-10-10 split: 80% for training (237 images), 10% for validation (30 images), and 10% for testing (29 images). This split ensures sufficient data for training while maintaining independent sets for validation and final performance evaluation. The fine-tuning process utilizes the Geotorch framework and runs for 10-30 epochs, with the exact number determined through monitoring validation performance to ensuring adequate model convergence, while keeping in mind the computational intensity of training a model of this scale.

4 Experimental setup

4.1 Data

For this research, we needed satellite data in HLS format that has Red, Green, Blue, Narrow NIR, SWIR 1 and SWIR 2 bands. The visible spectrum bands (Red, Green, and Blue) detect reflected light as seen by the human eye, enabling natural color imagery. The Narrow Near-Infrared (NIR) band is particularly sensitive to vegetation health and biomass, as healthy vegetation strongly reflects NIR radiation. The Short-Wave Infrared bands (SWIR1 and SWIR2) are crucial for burn severity assessment, as they can penetrate thin clouds and smoke, and are particularly effective at distinguishing between burned and unburned areas due to their sensitivity to moisture content and soil composition. These complementary bands, when analyzed together, provide comprehensive information about surface conditions and fire impacts that would be impossible to detect with visible light alone. In addition to that, we needed data that would include information on the burn severity in the form of a grading mask. For this reason, we chose to use the data provided by ibm-nasa-prithvi-models-family$_2$024, $which satisfies all of the se constrains. While the data provided there include images be fore the fire, during the fire and after the fire, we will be$ $fire scenario, as this is the only one relevant for our project.$

4.2 Model Architecture and Implementation

Our implementation builds upon the Prithvi architecture, specifically adapting it for burn intensity classification. Initially, we had several problems when setting up the processes due to the MMEngine framework, as previous versions of Prithvi had become outdated with respect to this dependency. We then established our training pipeline by integrating the Geotorch module provided by IBM designed specifically to aid in fine-tuning models such as Prithvi, creating a robust framework for our fine-tuning experiments. Consequently, we designed our model architecture to have the following structure:

(1) Base Architecture:
- Pretrained Prithvi backbone
- Semantic segmentation task configuration

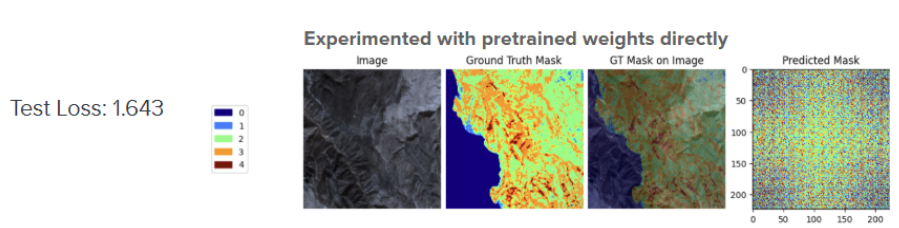
- Six input spectral bands: Blue, Green, Red, Narrow NIR, SWIR1, and SWIR2
- Classification output for 5 burn severity classes
- Cross Entropy loss function for multi-class segmentation

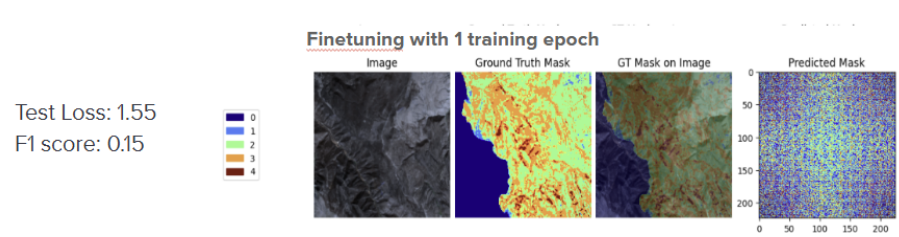
(2) Data Module:
- BurnIntensityNonGeoDataModule implementation
- Batch size of 8
- Weights initialization to address data imbalance issues

(3) Training Configuration:
- Training duration: 10 epochs (approximately 30 minutes per epoch)
- Validation performed every 2 epochs
- Optimizer: AdamW
- TensorBoardLogger implementation for training monitoring

5 Results

Our initial experiments comprised two distinct approaches to evaluate the model's performance in burn intensity classification. First, we tested the pre-trained Prithvi model directly on our dataset without any fine-tuning, which resulted in a test loss of 1.643. The predicted mask for this baseline experiment showed significant noise and poor correlation with the ground truth, indicating that the pre-trained weights alone were insufficient for accurate burn intensity classification. In our second experiment, we conducted minimal fine-tuning, training the model for a single epoch. This brief adaptation period yielded improved results, with the test loss decreasing to 1.55 and achieving an F1 score of 0.15. While this represents a modest improvement over the baseline, the predicted mask still shows considerable deviation from the ground truth, suggesting that more extensive training is necessary.





As shown on figures ?? and ?? both approaches struggle with accurate class differentiation across the five burn intensity levels (0-4). The ground truth masks show clear delineation between intensity zones, while our current predictions exhibit a more scattered, noise-like pattern.

The next phase of our experimentation will involve training the model for the full 10-30 epochs, which we anticipate will significantly improve the model's ability to distinguish between different burn intensity levels. This extended training period should allow the model to better learn the subtle spectral signatures associated with varying degrees of burn severity.

6 Conclusions

Our research demonstrates both the potential and challenges of fine-tuning the Prithvi model for burn intensity classification from satellite imagery. We successfully established a complete data pipeline for processing Harmonized Landsat Sentinel (HLS) data, creating a foundation for burn intensity analysis. The preliminary results show promising signs of learning capability, with observable improvements in accuracy even after minimal fine-tuning, suggesting the viability of our approach. However, several technical challenges emerged during this investigation. Working with Prithvi proved to be more complex than initially anticipated, primarily due to the immence computational capacity required to work with the model as well as due to the rapid changes in its supporting libraries. These continuous updates led to version compatibility issues, highlighting the challenges of working with cutting-edge deep learning frameworks in remote sensing applications. Moving forward, our findings indicate two clear directions for future work. First, extended training periods are necessary to fully realize the model's potential for burn intensity classification. Second, experimentation with different versions of Prithvi may yield insights into optimal model configurations for this specific task.

Received 20 February 2007; revised 12 March 2009; accepted 5 June 2009